Inputs: RegB (16 bits), RegC or Imm (as dictated by MUX, 16 bits)), Opcode (3 bits)

Output: Result (16 bits)

Implementation:

The ALU's functionality is dependent largely on the opcode input, which dictates the operation to be implemented.

ADD (000):

- Result stores the value of RegB + RegC_or_Imm

ADDI (001):

- Result stores the value of RegB + RegC_or_Imm

NAND (010):

- Result stores the value of RegB NAND RegC_or_Imm

LUI (011):

- Result stores the value of the lower 10 bits of RegC_or_Imm concatenated with 6 trailing zeros.

LW (100):

- Result stores the value of RegB + RegC_or_Imm, and serves as the memory address

SW (101):

- Result stores the value of RegB + RegC_or_Imm, and serves as the memory address

BEQ (110):

- Assuming RegB is holding the value of PC+1, Result stores RegB + RegC_or_Imm and serves as the memory address

JALR (111):

- ALU is not touched for this operation as PC+1 can be pulled directly from the appropriate wire.